

What Is Sovereign AI?

Sovereign AI refers to a nation's capabilities to produce artificial intelligence using its own infrastructure, data, workforce and business networks.

February 28, 2024 by [Angie Lee](#)



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Reading Time: 3 mins

Nations have long invested in domestic infrastructure to advance their economies, control their own data and take advantage of technology opportunities in areas such as transportation, communications, commerce, entertainment and healthcare.

AI, the most important technology of our time, is turbocharging innovation across every facet of society. It's expected to generate trillions of dollars in economic dividends and productivity gains.

Countries are investing in sovereign AI to develop and harness such benefits on their own. Sovereign AI refers to a nation's capabilities to produce artificial intelligence using its own infrastructure, data, workforce and business networks.

Why Sovereign AI Is Important

The global imperative for nations to invest in sovereign AI capabilities has grown since the rise of generative AI, which is reshaping markets, challenging governance models, inspiring new industries and transforming others — from gaming to biopharma. It's also rewriting the nature of work, as people in many fields start using AI-powered “copilots.”

Sovereign AI encompasses both physical and data infrastructures. The latter includes sovereign foundation models, such as large language models, developed by local teams and trained on local datasets to promote inclusiveness with specific dialects, cultures and practices.

For example, speech AI models can help preserve, promote and revitalize indigenous languages. And LLMs aren't just for teaching AIs human languages, but for writing software code, protecting consumers from financial fraud, teaching robots physical skills and much more.

In addition, as artificial intelligence and accelerated computing become increasingly critical tools for combating climate change, boosting energy efficiency and protecting against cybersecurity threats, sovereign AI has a pivotal role to play in equipping every nation to bolster its sustainability efforts.

Factoring In AI Factories

Comprising new, essential infrastructure for AI production are “AI factories,” where data comes in and intelligence comes out. These are next-generation data centers that host advanced, full-stack accelerated computing platforms for the most computationally intensive tasks.

Nations are building up domestic computing capacity through various models. Some are procuring and operating sovereign AI clouds in collaboration with state-owned telecommunications providers or utilities. Others are sponsoring local cloud partners to provide a shared AI computing platform for public- and private-sector use.



“The AI factory will become the bedrock of modern economies across the world,” NVIDIA founder and CEO Jensen Huang said in a recent media Q&A.

Sovereign AI Efforts Underway

Nations around the world are already investing in sovereign AI.

Since 2019, NVIDIA's AI Nations initiative has helped countries spanning every region of the globe to build sovereign AI capabilities, including ecosystem enablement and workforce development, creating the conditions for engineers, developers, scientists, entrepreneurs, creators and public sector officials to pursue their AI ambitions at home.

France-based Scaleway, a subsidiary of the iliad Group, is building Europe's most powerful cloud-native AI supercomputer. The NVIDIA DGX SuperPOD comprises 127 DGX H100 systems, representing 1,016 NVIDIA H100 Tensor Core GPUs interconnected by NVIDIA NVLink technology and the NVIDIA Quantum-2 InfiniBand platform. NVIDIA DGX systems also include NVIDIA AI Enterprise software for secure, supported and stable AI development and deployment.

Swisscom Group, majority-owned by the Swiss government, recently announced its Italian subsidiary, Fastweb, will build Italy's first and most powerful NVIDIA DGX-powered supercomputer — also using NVIDIA AI Enterprise software — to develop the first LLM natively trained in the Italian language.

With these NVIDIA technologies and its own cloud and cybersecurity infrastructures, Fastweb plans to launch an end-to-end system with which Italian companies, public-administration organizations and startups can develop generative AI applications for any industry.

The government of India has also announced sovereign AI initiatives promoting workforce development, sustainable computing and private-sector investment in domestic compute capacity. India-based [Tata Group](#), for example, is building a large-scale AI infrastructure powered by the [NVIDIA GH200 Grace Hopper Superchip](#), while [Reliance Industries](#) will develop a foundation LLM tailored for generative AI and trained on the diverse languages of the world's most populous nation. NVIDIA is also working with India's top universities to support and expand local researcher and developer communities.

Japan is going all in with sovereign AI, collaborating with NVIDIA to upskill its workforce, support Japanese language model development, and expand AI adoption for natural disaster response and climate resilience. These efforts include public-private partnerships that are incentivizing leaders like [SoftBank Corp.](#) to collaborate with NVIDIA on building a generative AI platform for 5G and 6G applications as well as a network of distributed AI factories.

Finally, Singapore is fostering a range of sovereign AI programs, including by partnering with NVIDIA to upgrade its National Super Computer Center, or NSCC, with NVIDIA H100 GPUs. In addition, [Singtel](#), a leading communications services provider building energy-efficient AI factories across Southeast Asia, is accelerated by [NVIDIA Hopper architecture GPUs](#) and NVIDIA AI reference architectures.

Read more about [sovereign AI and its transformative potential](#).

Explore [generative AI sessions and experiences at NVIDIA GTC](#), the global conference on AI and accelerated computing, running March 18-21 in San Jose, Calif., and online.



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NVIDIA CEO: Every Country Needs Sovereign AI

Jensen Huang describes transformative potential of sovereign AI at World Governments Summit.

February 12, 2024 by [Brian Caulfield](#)



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Reading Time: 3 mins

Every country needs to own the production of their own intelligence, NVIDIA founder and CEO Jensen Huang told attendees Monday at the World Governments Summit in Dubai.

Huang, who spoke as part of a fireside chat with the UAE's Minister of AI, His Excellency Omar Al Olama, described sovereign AI — which emphasizes a country's ownership over its data and the intelligence it produces — as an enormous opportunity for the world's leaders.

"It codifies your culture, your society's intelligence, your common sense, your history – you own your own data," Huang told Al Olama during their conversation, a highlight of an event attended by more than 4,000 delegates from 150 countries.

"We completely subscribe to that vision," Al Olama said. "That's why the UAE is moving aggressively on creating large language models and mobilizing compute."

Huang's appearance in the UAE comes as the Gulf State is moving rapidly to transform itself from an energy powerhouse into a global information technology hub.

Dubai is the latest stop for Huang in a global tour that has included meetings with leaders in Canada, France, India, Japan, Malaysia, Singapore and Vietnam over the past six months.

The Middle East is poised to reap significant benefits from AI, with PwC projecting a \$320 billion boost to the region's economy by 2030.

At Monday's summit, Huang urged leaders not to be "mystified" by AI. AI's unprecedented ability to take directions from ordinary humans makes it critical for countries to embrace AI, infusing it with local languages and expertise.

In response to AI Olama's question about how he might approach AI if he were the leader of a developing nation, Huang emphasized the importance of building infrastructure.

"It's not that costly, it is also not that hard," Huang said. "The first thing that I would do, of course, is I would codify the language, the data of your culture into your own large language model."

And as AI and accelerated computing has developed, NVIDIA GPUs have become a platform for one innovation after another.

"NVIDIA GPU is the only platform that's available to everybody on any platform," Huang said. "This ubiquity has not only democratized AI but facilitated a wave of innovation that spans from cloud computing to autonomous systems and beyond.

All of this promises to unleash new kinds of innovations that go beyond what's traditionally been thought of as information technology.

Huang even countered advice offered by many visionaries over the years who urged young people to study computer science in order to compete in the information age. No longer.



"In fact, it's almost exactly the opposite," Huang said. "It is our job to create computing technologies that nobody has to program and that the programming language is human: everybody in the world is now a programmer — that is the miracle."

In a move that further underscores the regional momentum behind AI, Moro Hub, a subsidiary of Digital DEWA, the digital arm of the Dubai Electricity and Water Authority, focused on providing cloud services, cybersecurity and smart city solutions, announced Monday it has agreed to build a green data center with NVIDIA.

In addition to the fireside chat, the summit featured panels on smart mobility, sustainable development and more, showcasing the latest in AI advancements. Later in the evening, Huang and AI Olama took the stage at the "Get Inspired" ecosystem event, organized by the UAE's AI Office, featuring 280 attendees including developers, startups and others.

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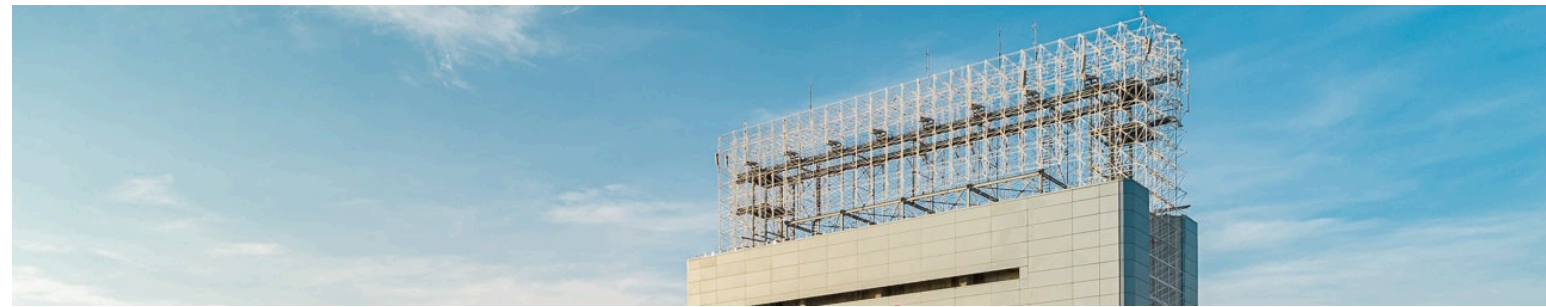
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Singtel, NVIDIA to Bring Sovereign AI to Southeast Asia

Singapore's leading service provider chooses the NVIDIA AI platform to deliver generative AI to its regional customers.

January 30, 2024 by [Joao Gomes](#)



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Asia's lion city is roaring ahead in AI.

Singtel, a leading communications services provider based in Singapore, will bring the [NVIDIA AI platform](#) to businesses in the island nation and beyond.

The mobile and broadband company is building energy-efficient data centers across Southeast Asia accelerated with [NVIDIA Hopper architecture GPUs](#) and using NVIDIA AI reference architectures proven to deliver optimal performance.

The data centers will serve as sovereign national resources — AI factories that process the private datasets of companies, startups, universities and governments safely on shore to produce valuable insights.

Singtel's first AI services will spin up in Singapore, with future data centers under construction in Indonesia and Thailand. From its hub in Singapore, the company has operations that stretch from Australia to India.

Trusted Engines of AI

The new data centers will act as trusted engines of generative AI. The most transformative technology of our time, generative AI and its ability to amplify human intelligence and productivity are attracting users worldwide.

Nations are creating large language models tuned to their local dialects, cultures and practices. Singtel sits at the center of such opportunities among Southeast Asia's vibrant Chinese, Indian, Malay and other communities.

Singtel's initiative supports Singapore's national AI strategy to empower its citizens with the latest technology. The plan calls for significantly expanding the country's compute infrastructure as well as its talent pool of machine learning specialists.

For businesses in the region, having a known, local provider of these computationally intensive services provides a safe, easy on-ramp to generative AI. They can enhance and personalize their products and services while protecting sensitive corporate data.

Taking the Green Path

Singtel is committed to democratizing AI and decarbonizing its operations.

Its latest data centers are being built with an eye to sustainability, including in the selection of materials  and use of liquid cooling. They adopt best practices to deliver less than 1.3 in PUE, the power usage effectiveness metric for data center efficiency.

Singtel will use its Paragon software platform to orchestrate how the new AI applications work in concert with its mobile and broadband services. The combination will enable edge computing services like powering robots and other autonomous systems from AI models running in the cloud.

A Full-Stack Foundation

The company will offer its customers NVIDIA AI Enterprise, a software platform for building and deploying AI applications, including generative AI. Singtel will also be an NVIDIA Cloud Partner, delivering optimized AI services on the NVIDIA platform.

Because Singtel's data centers use NVIDIA's proven reference architectures for AI computing, users can employ its services, knowing they're optimized for leading AI performance.

Singtel already has hands-on experience delivering edge services with NVIDIA AI.

Last May, it demonstrated a digital avatar created with the NVIDIA Omniverse and NVIDIA NeMo platforms that users could interact with over its 5G network. And in 2021, Singtel delivered GPU services as part of a testbed for local government agencies.

New AI Role for Telcos

Singapore's service provider joins pioneers in [France](#), [India](#), Italy and Switzerland deploying AI factories that deliver generative AI services with data sovereignty.

To learn more about how Singtel and other telcos are embracing generative AI, [register for a session](#) on the topic at [NVIDIA GTC](#). The global AI conference runs March 18-21, starting with a [keynote](#) by NVIDIA founder and CEO Jensen Huang.

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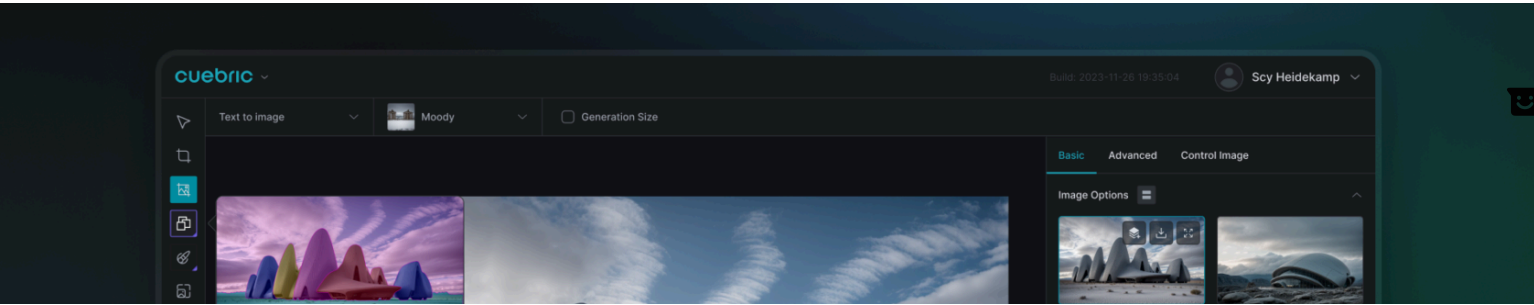
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And ... Action! Cuebric CEO Provides Insights Into Filmmaking Using AI

February 28, 2024 by [Brian Caulfield](#)



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Reading Time: 2 mins

These days, just about everyone is a content creator. But can generative AI help make people create high-quality films and other content affordably? Find out from Pinar Seyhan Demirdag, cofounder and CEO of [Cuebric](#), during his conversation with NVIDIA AI Podcast host Noah Kravitz.

Cuebric is on a mission to offer new solutions in filmmaking and content creation through immersive, two-and-a-half-dimensional cinematic environments. Its AI-powered application aims to help creators quickly bring their ideas to life, making high-quality production more accessible.

Demirdag discusses how Cuebric uses [generative AI](#) to enable the creation of engaging environments affordably. Listen in to find out about the current landscape of content creation, the role of AI in simplifying the creative process, and Cuebric's participation in NVIDIA's [GTC technology conference](#).

The AI Podcast · Exploring Filmmaking with Cuebric's AI: Insights from Pinar Seyhan Demirdag – Ep. 314

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Time Stamps:

1:15: Getting to know Pinar Seyhan Demirdag and Cuebric

2:30: The beginnings and goals of Cuebric

4:45: How Cuebric's AI application works for filmmakers

9:00: Advantages of AI in content creation

13:20: Making high-quality production budget-friendly

17:35: The future of AI in creative endeavors

22:00: Cuebric at NVIDIA GTC

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[UF Provost Joe Glover on Building a Leading AI University – Ep. 186](#)

Joe Glover, provost and senior vice president of academic affairs at the University of Florida, discusses the university's efforts to implement AI across all aspects of higher education, including a public-private partnership with NVIDIA that has helped transform UF into one of the leading AI universities in the country.

[NVIDIA's Marc Hamilton on Building the Cambridge-1 Supercomputer During a Pandemic – Ep. 137](#)

Cambridge-1, the U.K.'s most powerful supercomputer, ranks among the world's top 3 most energy-efficient supercomputers and was built to help healthcare researchers make new discoveries. Marc Hamilton, vice president of solutions architecture and engineering at NVIDIA, speaks on how he remotely oversaw its construction.

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